Course Outline

EN.705.744: 🤖 Deep Learning using Transformers Fall 2026

This outline provides an overview of the course and assignments by week. Please also check the Calendar in Canvas for specific due dates. Each course module runs for seven (7) days, i.e., one week, starting on the day of the first week (**Mondays**). Due dates for assignments are referred to by the day of the module week in which they are due. For example, if an Assignment is due by Day 7, then the Assignment should be completed by **Sunday** or the 7th day of the module.

Module	Dates	Module Title	Assignments
Module 1	Mon 08/31	**Foundations of Transformer Networks**	• Assignment 1 due • Discussion 1 start
Module 2	Mon 09/07	**Large Language Models LLM** Transformer Scaling Laws	• Assignment 2 due • Research Paper pick topic
Module 3	Mon 09/14	**Vision Transformers** ViT	• Assignment 3 due
Module 4	Mon 09/21	**Hierarchical Vision Transformers** Swin, Pyramid	• Assignment 4 due • Discussion 2 start
Module 5	Mon 09/28	**Object Detection** DETR, Deformable Attention	• Assignment 5 due
Module 6	Mon 10/05	**Semantic Segmentation** SETR, MaskFormer	• Assignment 6 due
Module 7	Mon 10/12	**Video Understanding** ViViT, VisTR	• Assignment 7 due • Discussion 3 start
Module 8	Mon 10/19	**Contrastive Learning, Mask Img Modeling** MAE, SimMIM	• Assignment 8 due • Research Paper draft
Module 9	Mon 10/26	**Self-Supervised Learning** DINO	• Assignment 9 due
Module 10	Mon 11/02	**Multimodal Learning** CLIP	• Assignment 10 due • Discussion 4 start
Module 11	Mon 11/09	**3D, Point Cloud** Point and Graph Transformers	• Assignment 11 due
Module 12	Mon 11/16	**Parameter Efficient Fine Tuning** LoRA	• Assignment 12 due
11/23–11/29 - Thanksgiving break - Nothing assigned or due			
Module 13	Mon 11/30	**Hybrid CNN-Transformer Architecture** UniFormer	• Assignment 13 due
Module 14	Mon 12/07	**Foundation Models** Mixture of Experts	• Assignment 14 due • Research Paper due